

Data Warehouses and Data Mining Systems

Report no. 2

Exercise topic:

Phase 2: Conceptual modeling

Project topic:

Counter Strike skin market analysis

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Gliwice, 23.03.2026

1 Operational Database Model

The operational database model is designed to support transactional storage and efficient querying of both sticker and weapon skin market data.

1.1 Logical Model Description

The system consists of the following entities:

- Teams
- Matches
- Stickers
- Weapon Skins
- Weapons
- Cases
- Prices (time-series)

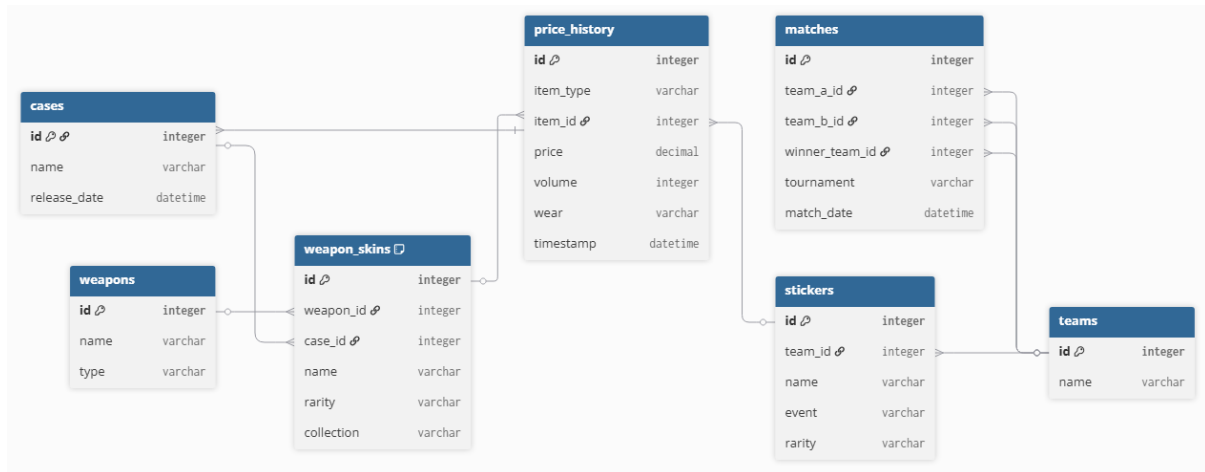
Key Attributes:

- Weapon Skin: rarity, wear condition
- Case: case price
- Weapon: weapon type

Relationships:

- One Team $\rightarrow$ many Stickers
- One Sticker $\rightarrow$ many Price records
- One Weapon $\rightarrow$ many Weapon Skins
- One Case $\rightarrow$ many Weapon Skins
- One Weapon Skin $\rightarrow$ many Price records
- One Match $\rightarrow$ one winning Team

1.2 ERD



1.3 Key Design Notes

- Time-dependent data (prices) is separated into a dedicated table
- Stickers are linked to teams for esports impact analysis
- Weapon skins are modeled independently with attributes affecting price:
 - rarity
 - wear condition
 - case origin (case price influence)
 - weapon type
- Case price is included as an economic driver of skin value

2 Data Warehouse Conceptual Model

The data warehouse is designed using a **star schema** optimized for analytical queries.

2.1 Fact Identification

Fact: Market Price

2.1.1 Measures

- price
- volume

2.2 Dimensions

- Time
- Item (Sticker / Weapon Skin)
- Team
- Weapon
- Case
- Match Outcome

2.3 Attribute Classification

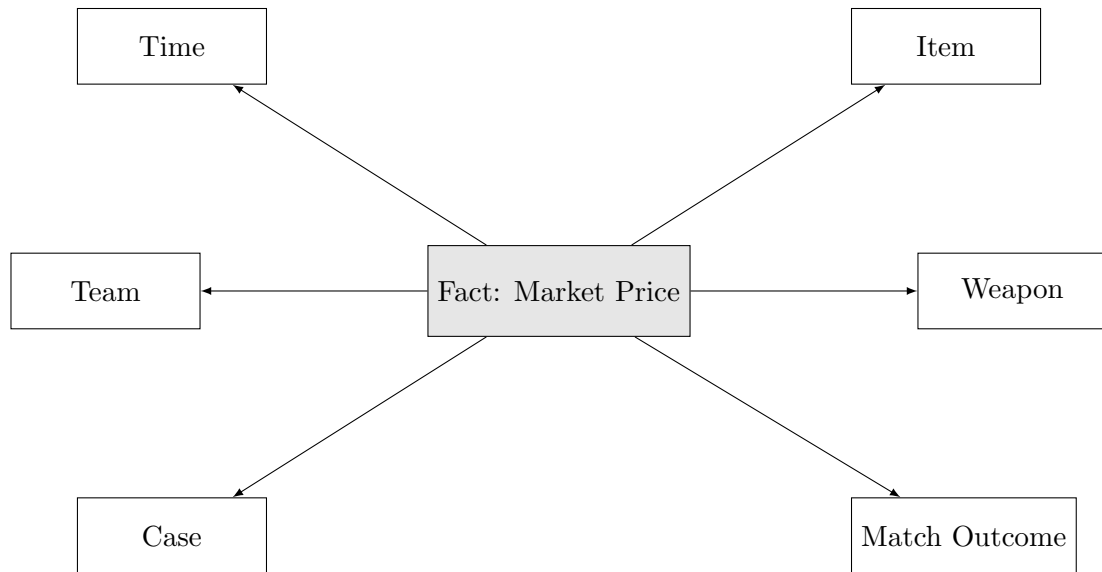
Measures:

- price
- volume

Dimensional Attributes:

- item type (sticker / skin)
- skin rarity, wear condition
- weapon type
- case price
- team name, region
- timestamp (day, month, year)
- match result (winner, loser, date)

2.4 Star Schema Diagram



2.5 Schema Interpretation

- Centralized fact table enables aggregation across all market items
- Time dimension supports trend and event-based analysis
- Item dimension unifies stickers and weapon skins
- Weapon, rarity, wear, and case introduce intrinsic price drivers
- Match outcomes enable analysis of esports-driven price changes

3 Conclusion

The operational model captures both esports-driven and intrinsic market factors affecting item prices. Weapon skins are modeled with attributes such as rarity, wear condition, case origin, and weapon type, enabling deeper economic analysis. The data warehouse integrates these factors into a star schema that supports both causal (match outcome) and structural (item characteristics) analysis of price behavior.